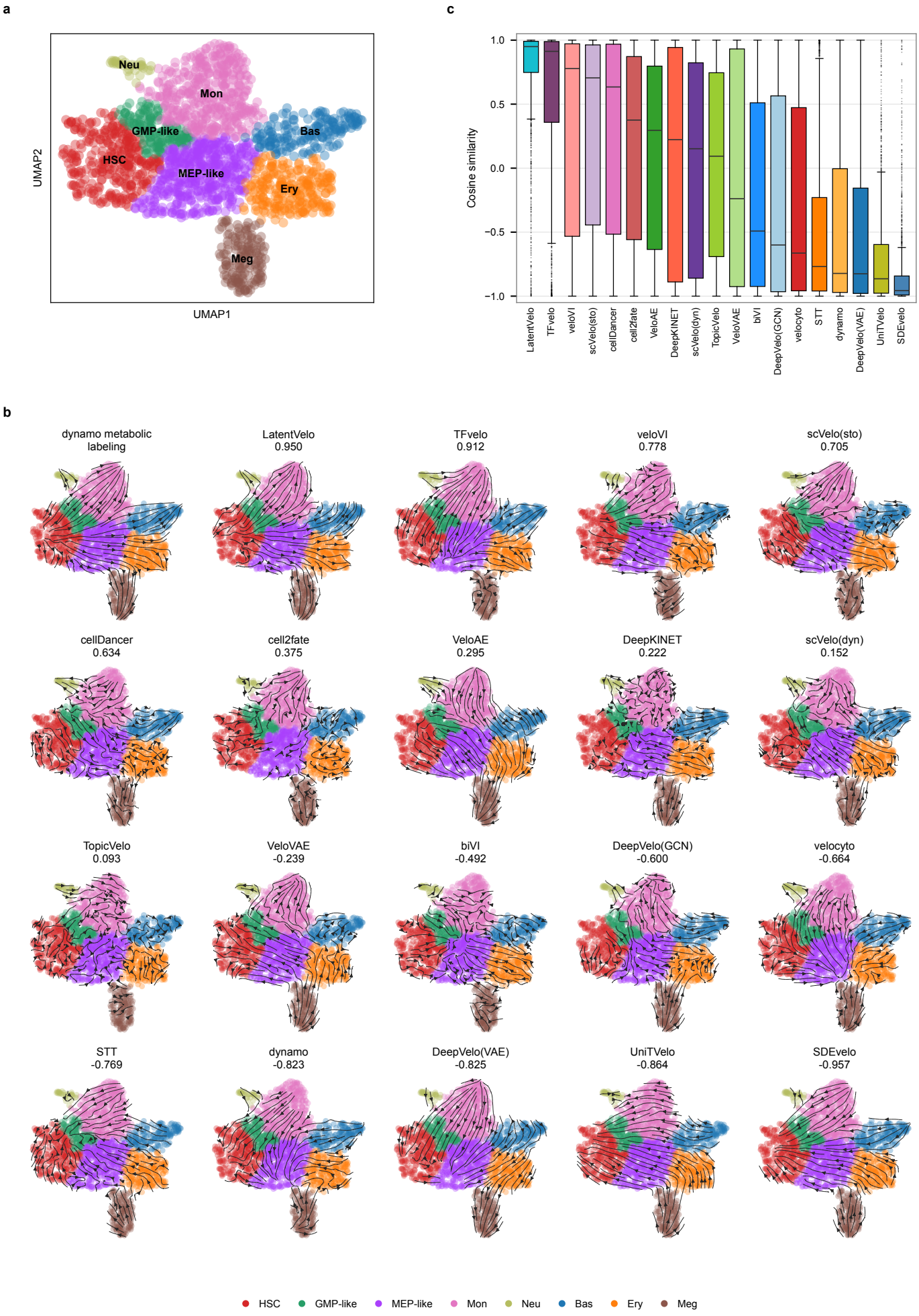

a
Neu
Mon
GMP-like
Bas
HSC
MEP-like
Ery
Meg
UMAP2
UMAP1
c
Cosine similarity
1.0
0.5
0.0
−0.5
−1.0
LatentVelo
TFvelo
veloVI
scVelo(sto)
cellDancer
cell2fate
VeloAE
DeepKINET
scVelo(dyn)
TopicVelo
VeloVAE
biVI
DeepVelo(GCN)
velocyto
STT
dynamo
DeepVelo(VAE)
UniTVelo
SDEvelo
b
dynamo metabolic labeling
LatentVelo 0.950
TFvelo 0.912
veloVI 0.778
scVelo(sto) 0.705
cellDancer 0.634
cell2fate 0.375
VeloAE 0.295
DeepKINET 0.222
scVelo(dyn) 0.152
TopicVelo 0.093
VeloVAE -0.239
biVI -0.492
DeepVelo(GCN) -0.600
velocyto -0.664
STT -0.769
dynamo -0.823
DeepVelo(VAE) -0.825
UniTVelo -0.864
SDEvelo -0.957
HSC
GMP-like
MEP-like
Mon
Neu
Bas
Ery
Meg

Extended Data Fig. 7 | Metabolic-labeling reference: scNT-seq. a, UMAP annotation of 1,947 scNT-seq cells, colored by cell type. **b,** Velocity streams for the dynamo metabolic-labeling reference and all 19 method configurations, arranged in five columns. **c,** Per-cell cosine similarity to the metabolic-labeling reference. Numbers below method names in the stream panels are the median per-cell reference-direction cosines, shown to three decimal places. Boxes show the median and interquartile range, with 1.5-IQR whiskers and outliers; methods are ordered by median. All 19 configurations use barcode-aligned cells and the same reference coordinates. Cosines were calculated from the displayed per-cell directions.